**POKHARA UNIVERSITY**

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Data Structure and Algorithm | Pass Marks : 45
Time : 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

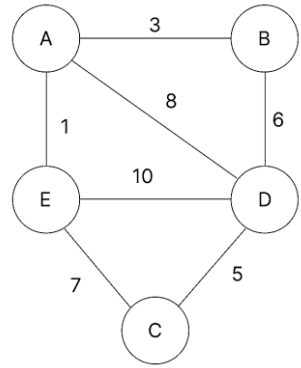
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|---|---|---|---|
| 1. | a) | What is abstract data type (ADT)? Explain the role of data structure in computer science. | 7 |
| | b) | Evaluate the expression ABCD-*+ using stack with A = 5, B = 4, C = 3 and D = 7. | 8 |
| 2. | a) | Implement enqueue and dequeue operations in circular queue using C/C++ language. | 8 |
| | b) | Define a linked list. Write an algorithm to insert a node at the end. | 7 |
| 3. | a) | Write an algorithm to insert a node at the beginning of the doubly circular linked list. | 7 |
| | b) | Explain the base case and recursive case in the recursive algorithm for solving the Tower of Bhramha. | 8 |
| 4. | a) | Define a complete binary tree. Why do you need to balance a binary search tree? Explain with example. | 7 |
| | b) | Construct an AVL tree from following sequence of numbers inserted in an order: 30,11,5,22,17,18,50,45 | 8 |
| 5. | a) | Define internal sorting. Sort the data 40,30,12,21,4,5,78,6,5 using Insertion Sort. | 7 |
| | b) | What is Hashing? With given input {4371, 1323, 1222, 3424, 6173, 4199, 4344, 9679, 1989} and hash function: h(x) = x mod 10, show the following. | 8 |

i. Hash table using linear probing
ii. Hash table using quadratic probing



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6. a) Define directed graph. Find minimum spanning tree of following graph using Kruskal's Algorithm. 7




b) When do we use Breadth First Search and Depth First Search in graph problem? Explain any one with example. 8

7. Write short notes on: (**Any two**) 2×5

a) Push and POP operations

b) Divide and Conquer algorithm

c) B Tree

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